

Sol. 1) Let, x_1 = Number of model-1

x_2 = Number of model-2

Objective function:

$$\text{Maximize } Z = 30x_1 + 40x_2$$

Subject to:

$$2x_1 + 3x_2 \leq 1200$$

$$2x_1 + x_2 \leq 1000$$

$$4x_2 \leq 800$$

$$x_1 \geq 0, x_2 \geq 0$$

Augmented problem: -

$$\text{Maximize } Z = 30x_1 + 40x_2 + 0S_1 + 0S_2 + 0S_3$$

S.T:

$$2x_1 + 3x_2 + S_1 = 1200$$

$$2x_1 + x_2 + S_2 = 1000$$

$$4x_2 + S_3 = 800$$

$$x_i \geq 0 \quad \forall i = 1 \dots 5.$$

Iteration 0

		C_j	30	40	0	0	0
C_j	Basis	Value	x_1	x_2	S_1	S_2	S_3
0	S_1	1200	2	3	1	0	0
0	S_2	1000	2	1	0	1	0
0	S_3	800	0	4	0	0	1
		$C_j - Z_j$	30	40	0	0	0

Iteration 1

		C_j	30	40	0	0	0
C_j	Basis	Value	x_1	x_2	S_1	S_2	S_3
0	S_1	600	2	0	1	0	-3/4
0	S_2	800	2	0	0	1	-1/4
40	x_2	200	0	1	0	0	1/4
		$C_j - Z_j$	30	0	0	0	-10

Iteration 2

		C_j	30	40	0	0	0
C_j	Basis	Value	x_1	x_2	S_1	S_2	S_3
30	x_1	450	1	0	-1/4	3/4	0
0	S_2	400	0	0	-2	2	1
40	x_2	100	0	1	1/2	-1/2	0
		$C_j - Z_j$	0	0	-25/2	-5/2	0

Optimal solution: $x_1^* = 450$; $x_2^* = 100$; $Z = 17,500$

(b) Basis matrix (B): - This is the matrix formed by the variables in the basis of the final table. Its values, however, will be obtained from the initial table.

$$B = \begin{bmatrix} 2 & 0 & 3 \\ 2 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}$$

Inverse of the basis matrix (B^{-1}): This matrix is formed by the variables in the basis of initial table and values from final table.

$$B^{-1} = \begin{bmatrix} -1/4 & 3/4 & 0 \\ -2 & 2 & 1 \\ 1/2 & -1/2 & 0 \end{bmatrix}$$

Resource (b_1) of milling machine changes from 1200 to 1300.

Changes in resource matrix affect the optimal solution. Therefore, new optimal solution will be: -

$$B^{-1}b = \begin{bmatrix} -1/4 & 3/4 & 0 \\ -2 & 2 & 1 \\ 1/2 & -1/2 & 0 \end{bmatrix} \begin{bmatrix} 1300 \\ 1000 \\ 800 \end{bmatrix} = \begin{bmatrix} 425 \\ 200 \\ 150 \end{bmatrix}$$

$x_1^* = 425$; $x_2^* = 150$; $Z = 18,750$

(C) Here, Resource (b_2) of Grinding machine changes from 800 to 350

$$B^{-1}b = \begin{bmatrix} -1/4 & 3/4 & 0 \\ -2 & 2 & 1 \\ 1/2 & -1/2 & 0 \end{bmatrix} \begin{bmatrix} 1200 \\ 1000 \\ 350 \end{bmatrix} = \begin{bmatrix} 450 \\ -50 \\ 100 \end{bmatrix}$$

No, we cannot determine the new optimal solution directly from the given information. Because $S_3 \leq 0$ i.e. infeasible.

Sol 2) Dual problem

Minimize $z = 5y_1 + 3y_2 + 8y_3$

Subject to $y_1 - y_2 + 4y_3 = 5$
 $2y_1 + 5y_2 + 7y_3 \geq 6$

y_1 unrestricted, $y_2 \leq 0$, $y_3 \geq 0$

Sol 3) Maximize $2x_1 + x_2$

subject to $x_1 \leq 0$

let, $x_1 = -x_1'$; $x_2 = x_2^+ - x_2^-$

Now the problem transforms into:

Maximize $-2x_1' + x_2^+ - x_2^-$

Subject to: $x_1' \geq 0$

$x_2^+ \geq 0$

$x_2^- \geq 0$

- (a) The dual form does not exist.
- (b) Dual solution is infeasible.
- (c) Primal optimal solution cannot be obtained because dual form is infeasible.

Sol 4) From the table, the starting primal variable x_4 and R uniquely correspond to the dual variables y_1 and y_2 , respectively. Thus, we determine the optimum dual solution as follows:

Starting primal basic variables	x_4	R
Z- equation coefficients	29/5	-2/5+M
Original objective coefficient	0	-M
Dual variables	y_1	y_2
Optimal dual values	29/5+0=29/5	-2/5+M+(-M) = -2/5

5) Minimize $Z = 600x_1 + 500x_2$
s.t. $2x_1 + x_2 \geq 80$
 $x_1 + 2x_2 \geq 60$
 $x_1, x_2 \geq 0$

Dual:
Maximize $w = 80y_1 + 60y_2$
s.t. $2y_1 + y_2 \leq 600$
 $y_1 + 2y_2 \leq 500$

	x_1	x_2	y_3	y_4	b	ratio
x_3	2	1	1	0	600	300
y_4	1	2	0	1	500	500
$(R_j - C_j)$	-80	-60	0	0	0	
x_1	1	1/2	1/2	0	300	600
y_4	0	3/2	-1/2	1	200	400/3
$(Z_j - C_j)$	0	-20	40	0	24000	
x_1	1	0	2/3	-1/3	700/3	
y_2	0	1	-1/3	2/3	400/3	
	0	0	100/3	40/3	80000/3	

$x_1 = \frac{700}{3}$, $x_2 = \frac{400}{3}$, $w = \frac{80,000}{3}$

6) Minimize

$$Z = x_1 + 2x_2 + 3x_3$$

s.t.

$$3x_1 + 4x_2 \leq 5$$

$$5x_1 + x_2 + 6x_3 = 7$$

$$8x_1 + 9x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0$$

Dual:

$$\text{Maximize } W = 5y_1 + 7y_2 + 2y_3$$

$$\text{s.t.}, 3y_1 + 5y_2 + 8y_3 \geq 1$$

$$4y_1 + y_2 = 2$$

$$6y_2 + 9y_3 \leq 3$$

	y_1	y_2	y_3	y_4	y_5	Δ_1	Δ_2	b	ratio
y_5	0	6	9	0	1	0	0	3	$1/3$
x_1	3	5	8	-1	0	1	0	1	$1/8$
x_2	4	1	0	0	0	0	1	2	-
Z	-5	-7	-2	0	0	0	0	0	
Z'	-7	-6	-8	1	0	0	0	-3	
y_5	0	6	9	0	1	0	0	3	-
y_1	1	$5/3$	$8/3$	$-1/3$	0	$1/3$	0	$1/3$	-
x_2	0	$-17/3$	$-32/3$	$4/3$	0	$-4/3$	1	$2/3$	$1/2$
Z	0	$4/3$	$34/3$	$-5/3$	0	$5/3$	0	$5/3$	
Z'	0	$17/3$	$32/3$	$-4/3$	0	$7/3$	0	$-2/3$	
y_5	0	6	9	0	1	0	0	3	$1/2$
y_1	1	$1/4$	0	0	0	0	$1/4$	$1/2$	2
y_4	0	$-17/4$	-8	1	0	-1	$3/4$	$1/2$	-
Z	0	$-23/4$	-2	0	0	0	$5/4$	$5/2$	
Z'	0	0	0	0	0	0	1	0	
y_2	0	1	$3/2$	0	$1/6$	$-1/24$		$1/2$	
y_1	1	0	$-3/8$	0	$13/24$			$3/8$	
y_4	0	0	$-13/8$	1				$21/8$	
Z	0	0	$53/8$	0	$23/24$			$43/8$	

$$y_1 = 3/8$$

$$y_2 = 1/2$$

$$y_4 = 21/8$$

$$W = 43/8$$

7.

BV	Eq. #	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	1	1	1	0	0	0	0
x_3	1	0	1	1	1	0	0	8
x_4	2	0	0	-1*	0	1	0	-3
x_5	3	0	-1	1	0	0	1	2
Z	0	1	1	0	0	1	0	-3
x_3	1	0	1	0	1	1	0	5
x_2	2	0	0	-1	0	-1	0	3
x_5	3	0	-1*	0	0	1	1	-1
Z	0	1	0	0	0	2	1	-4
x_3	1	0	0	0	1	2	1	4
x_2	2	0	0	1	0	-1	0	3
x_1	3	0	1	0	0	-1	-1	1

So $(x_1, x_2) = (1, 3)$ with $Z = -4$ is optimal